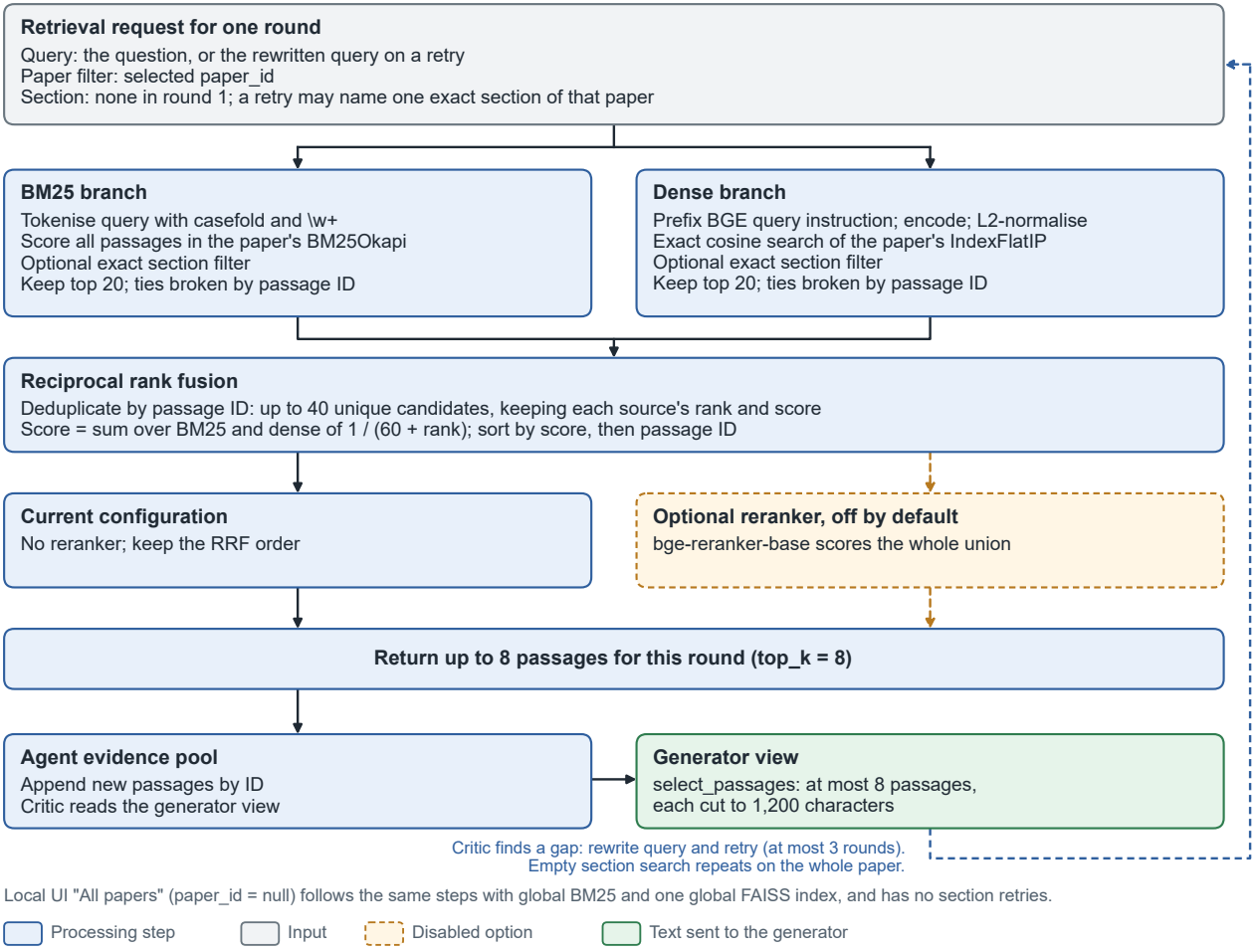

Retrieval request for one round
Query: the question, or the rewritten query on a retry
Paper filter: selected paper_id
Section: none in round 1; a retry may name one exact section of that paper
BM25 branch
Tokenise query with casefold and \w+
Score all passages in the paper's BM25Okapi
Optional exact section filter
Keep top 20; ties broken by passage ID
Dense branch
Prefix BGE query instruction; encode; L2-normalise
Exact cosine search of the paper's IndexFlatIP
Optional exact section filter
Keep top 20; ties broken by passage ID
Reciprocal rank fusion
Deduplicate by passage ID: up to 40 unique candidates, keeping each source's rank and score
Score = sum over BM25 and dense of 1 / (60 + rank); sort by score, then passage ID
Current configuration
No reranker; keep the RRF order
Optional reranker, off by default
bge-reranker-base scores the whole union
Return up to 8 passages for this round (top_k = 8)
Agent evidence pool
Append new passages by ID
Critic reads the generator view
Generator view
select_passages: at most 8 passages,
each cut to 1,200 characters
Critic finds a gap: rewrite query and retry (at most 3 rounds).
Empty section search repeats on the whole paper.
Local UI "All papers" (paper_id = null) follows the same steps with global BM25 and one global FAISS index, and has no section retries.
Processing step
Input
Disabled option
Text sent to the generator